A traced warp becomes a causal event path

One worked slice exposes information that a generic architecture block cannot show

(a) Observed trace + static control

warp 07 • block 12 • SM 3 • SMSP 2

one record = one warp instruction (32 lanes max)

record	SASS / pipe	lane evidence
r_1	LDG.E	mask 0xffff • S0–S3
r_2	HMMA	tensor • no memory
r_3	STG.E	mask 0x0fff • S4–S5

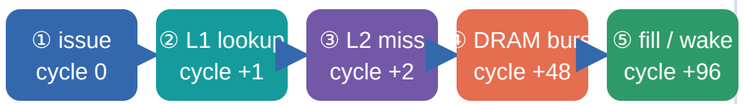
SASS control word

stall 4 • yield 0 • wait mask 0

read/write barriers + reuse bits are joined by PC

(b) Next-event causal path

same line fill merges two requests before DRAM



event-driven jump

clock 1 → 48 without polling idle cycles

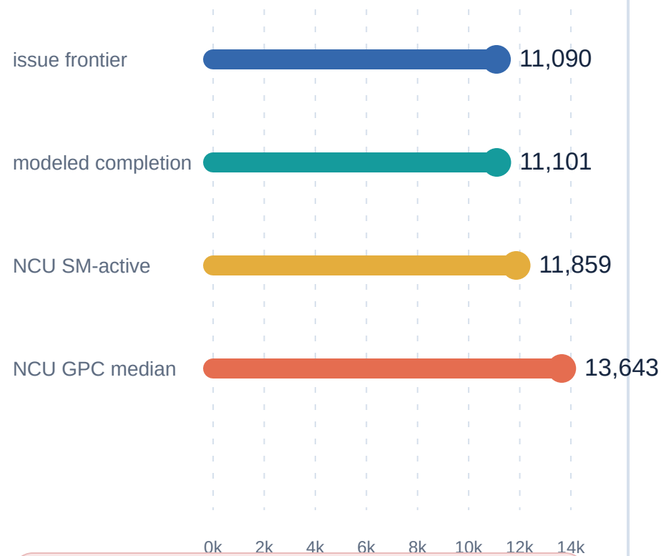
causal order is retained; empty time is not simulated

worked slice: 196,608 load sectors • 8,192 DRAM sectors

(c) Separate frontiers

gemm_hmma_bf16_sm80 • SM86

post-merge simulator / fresh NCU medians



typed tail boundary

coherence • dirty writeback

not silently folded into elapsed time

Legend

trace/data · modeled event · hardware observation · typed / diagnostic

Caption. Panels (a–c) follow one traced warp instruction through lane-aware decode, L1/L2/DRAM events, an MSHR merge, and separate issue/completion/observation frontiers. Numbers are the SM86 gemm_hmma_bf16_sm80 campaign row; the event slice is illustrative and does not invent a cycle-accuracy claim.

Source: Post-merge SM86 campaign manifests; sim-trace/sim-sm/sim-mem; docs/postmerge_campaign_comparison_20260909.md.